

AI DEVELOPMENT CAPACITY & DESIGN GENERATION ENGINE

ENVELOPE

What can be built on this plot —
answered in minutes, with every number traceable.



Dubai residential towers · Phase 0 to Phase 5 · Confidential



THE PROBLEM

The most expensive question in development is answered badly

“What can we build on this plot?” decides whether land is bought and at what price. It is currently answered slowly, once, and without evidence.

2–6 weeks

to a first capacity answer, by which time the bid deadline has usually passed

AED 50k–250k

per feasibility study, which makes exploring a second option uneconomic

One option

explored out of hundreds that were legally and commercially possible

Zero citations

behind the numbers that then drive the entire financial model

An efficiency percentage typed into a spreadsheet becomes a fact in the pro forma. Nobody can trace where it came from.

The rules already exist. They have never been executable.

Dubai has an unusually complete and consolidated regulatory framework. It is written for human readers — which is exactly why it can be structured.

Dubai Building Code

843 pages

Parts A–K. Defines FAR, gross floor area, setbacks, corridors, accessibility, vertical transportation.

UAE Fire & Life Safety Code

707 pages

Egress, travel distance, dead ends, exit capacity. Cross-referenced directly by the DBC.

Development Control Regulations

Per plot

Height, land use, setbacks and building lines for the specific plot. Overrides the code where it speaks.

Developer standards & RFPs

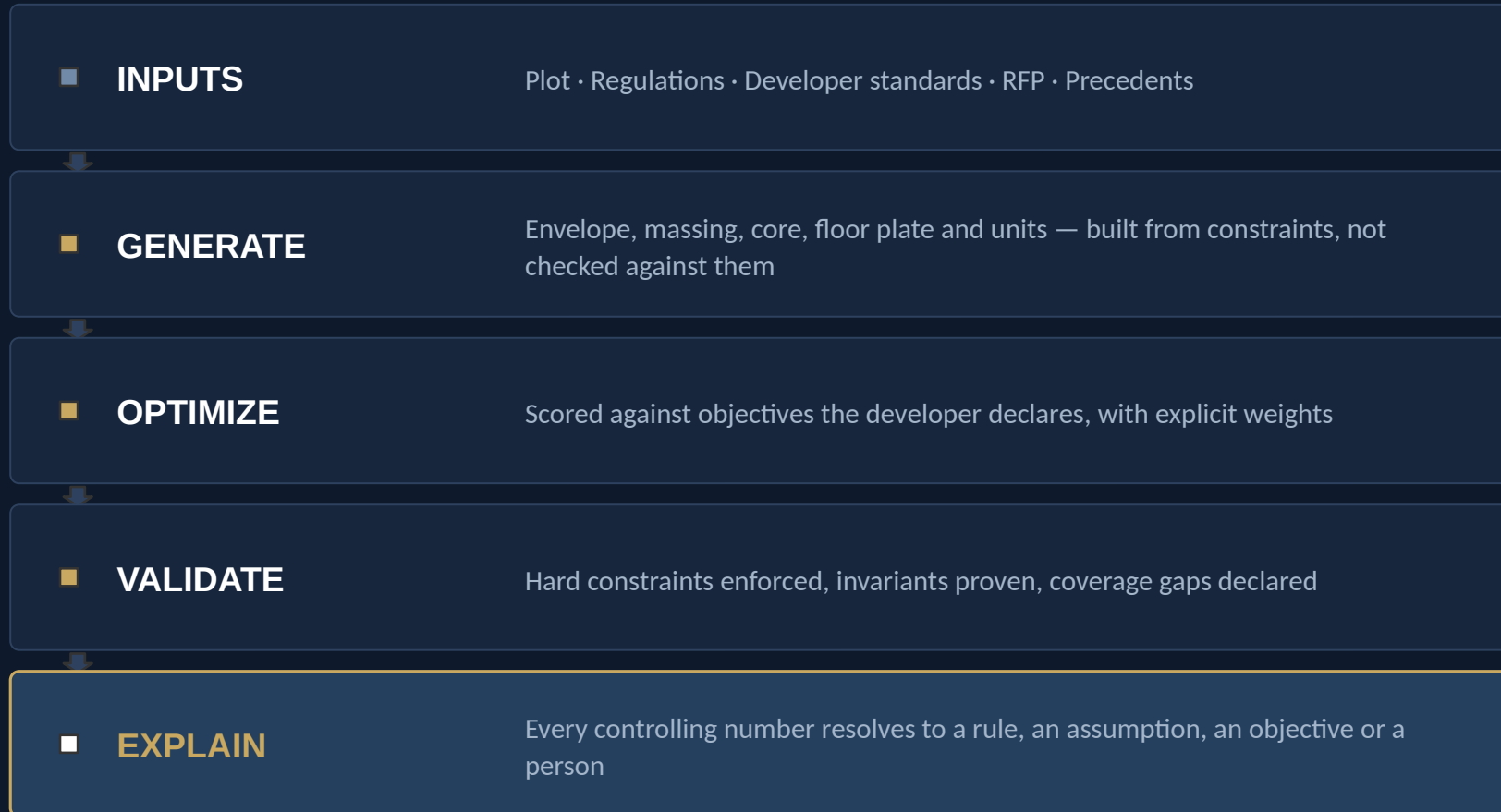
Per client

Unit mix, minimum areas, core and corridor preferences, efficiency targets, commercial objectives.

1,550 pages of shared rules — plus a different DCR for every plot.

No consultant reads all of it on any project. They apply the subset they remember. Different consultants apply different subsets, and nobody can prove which subset was used. Structuring this once, properly, is the asset.

From plot to development decision, in one pipeline



The order matters

Most tools start at the middle of this pipeline, because they need a design to already exist.

ENVELOPE starts at the top, with a plot and an intention, and nothing drawn.

That is the whole difference — and it is why validation sits inside generation rather than after it.

Precedents inform the answer. They never authorise it — a previously approved scheme is evidence, not permission.

A rule is not a test. It is an instruction.

A compliance checker asks whether a drawing passes. ENVELOPE turns the same rule into something that constructs buildable volume from an empty plot.

COMPLIANCE CHECKER

“Is this setback at least 5.25 m?”

Needs a drawing to exist.

Returns pass or fail.

Cannot tell you what to build instead.

ENVELOPE

“Offset every boundary inward by 5.25 m.”

Needs only a plot.

Returns the buildable footprint.

The rule generates the answer.

And the real code is harder than that

In the Dubai Building Code, the setback from a neighbouring plot depends on how many floors the building has — 3.00 m at ground, rising to 7.50 m at G+9 and above. But the number of floors depends on the footprint, and the footprint depends on the setback.

Assume G+10 setback 7.50 m → 7.6 floors

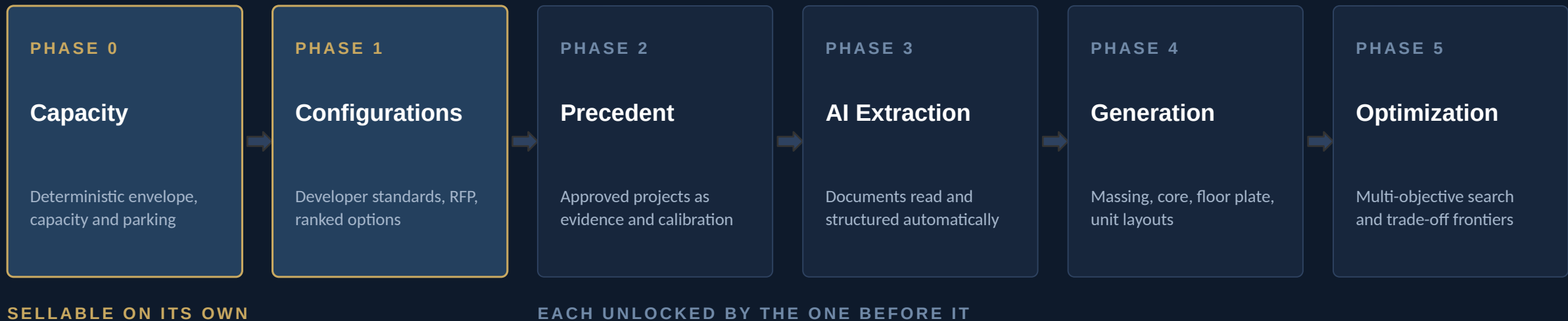
Assume G+7 setback 6.00 m → 6.9 floors

Assume G+6 setback 5.25 m → **6.6 floors ✓**

Converges in three passes. A checker cannot resolve this — it has nothing to check yet.

Six phases, each one a dependency of the next

The ordering is technical necessity, not caution. You cannot judge a generated tower until you can compute exactly what volume it may occupy.



- Phase 2 needs a deterministic baseline to calibrate precedent against.
- Phase 3 needs a rule model for extracted documents to be structured into.
- Phase 4 needs constraints that can construct, not merely check.
- Phase 5 needs a generator whose output is already trustworthy.

A deterministic engine that computes capacity and shows its working

One community, one land use, residential towers. Form-based input. No AI in the computation path — deliberately.

WHAT IT COMPUTES

- Buildable envelope from plot geometry
- FAR, height, coverage and setbacks applied
- Parking demand, area and podium implication
- Capacity bands, with the binding limit named
- Assumption register, ranked by sensitivity
- Invariant and constraint checks
- PDF report and JSON export

WHY NO AI IN PHASE 0

Extraction is not the risky part of this product. Geometry, rules and defensible provenance are.

- A form takes ninety seconds. The extraction pipeline takes six weeks to debug.
- AI extraction needs a rule model to extract into — Phase 0 builds it.
- It isolates the variable: if Phase 0 fails, we know it failed on geometry or rules, not on document reading.

8–12 weeks · four people · ten real plots · ≈40 rules · benchmarked against two independent architects

Speed is worthless if the number cannot be defended

Four mechanisms, all built in Phase 0, because none of them can be retrofitted later.

Provenance

Every number resolves to a cited rule, a declared assumption, an objective, or a named person. The system never says “the AI decided.”

Human-approved rules

AI may draft a rule. A licensed architect approves it. Unapproved rules cannot be loaded by the engine — there is no override flag.

Assumption register

Where no rule governs, the system declares an assumption, ranks it by how much it moves the answer, and requires acknowledgement before export.

Invariant checks

Independent arithmetic proofs: areas must sum, ratios must reconcile, parking must fit. A configuration that fails is never emitted.

No hidden defaults. If we filled a gap, it is amber on the screen and listed in the report.

Five capacities. Most feasibility work reports one.

Knowing which limit binds is worth more than knowing the maximum — it tells you whether to negotiate, redesign, or accept.

■ Regulatory capacity	What FAR and the area caps permit	Phase 0
■ Geometric capacity	What the envelope physically holds within height and footprint limits	Phase 0
■ Parking capacity	What the achievable parking supply can actually support	Phase 0
■ Commercially preferred capacity	What the developer's own declared objectives select	Phase 1
■ Historically observed capacity	What comparable approved plots actually achieved	Phase 2

The governing capacity is the smallest of them — and we name it, rather than reporting a maximum the plot will never deliver.

AI reads. The engine computes. Humans approve.

Language models are excellent at reading documents and writing explanations, and unacceptable at computing an area. We use each for what it is reliable at.

AI READS AND STRUCTURES

- Reads regulations, standards and RFPs
- Proposes structured candidate rules
- Classifies and normalises requirements
- Drafts explanations from the evidence

THE ENGINE COMPUTES

- Geometry, areas and envelopes
- Rule applicability and evaluation
- Capacity, parking and metrics
- Constraint enforcement and invariants

HUMANS APPROVE

- Licensed architects approve every rule
- Objectives and weights are declared, not inferred
- Assumptions acknowledged before export
- Final professional certification

An AI may draft a rule. It never becomes one. And it never computes a number you see on a report.

Four assets. The last one cannot be bought.

Ordered by how hard each is to replicate. The visible parts of the product are the least defensible.

01	Executable Dubai regulation model	Structured, cited, versioned, human-approved rules. Months of licensed-architect work, and every regulatory change we absorb widens the gap for a new entrant.	Hard
02	Developer standards library	Each client's house rules structured once, reused on every project. Creates switching cost from the second project onward.	Harder
03	Precedent corpus	Approved projects normalised into evidence — what capacity was actually achieved, not merely permitted.	Harder
04	Approval feedback loop	Which ENVELOPE-derived schemes were approved, amended or rejected. Nobody can purchase this dataset. It compounds with every project.	Hardest

Not a moat: the model, the interface, the optimiser. All commodity. We are not betting on any of them.

Phase 0 is not a calculator.

It is the foundation asset for the entire AI design-generation platform.

The rule model

Every later phase computes against it

The geometry engine

Generation and optimisation both run on it

The provenance system

Cannot be retrofitted — only rebuilt

The invariant layer

Makes an impossible building unpublishable

Build the Phase 0 engine. Validate it against real projects. Then scale it into the platform.